EGRE 245 Engineering Programming
Spring 2026
Mark Schwitzerlett

Lab Exercise #5: Olympic Scoring

1 Overview

In many Olympic events, multiple judges score a performance. To reduce bias, the highest and lowest scores are often removed, and the remaining scores are averaged.

Your lab today will simulate that scoring.

Write a C program that

1. Prompts the user to enter the number of judges (**between 5 and 10**).
2. Uses a one-dimensional array to store each judge score.
3. Prompts the user to enter each score as a decimal number.
4. Validates scores are between **0.0 and 10.0** (inclusive).
5. Computes and displays:
    - All scores entered (in the order entered)
    - The highest score
    - The lowest score
    - The **official score**:
        - Remove exactly one highest score and exactly one lowest score
        - Average the remaining scores
6. Use a 1D array.
7. Use at least one loop for input and one loop for analysis.
8. Do not use functions (everything inside `main`).
9. Use `printf` and `scanf` only for I/O.
10. Your output must include:
    - The list of judge scores
    - Highest and lowest values

- The official score formatted to **two decimal places**

2 Code Documentation

The very first lines in *all* of your programming project files this semester should be comments in the class header file (copy from header.c). Of course, the items in the square brackets [] in the comments should be filled in with the proper information.

In addition, the first output for all of your projects this semester, both for the lab exercises and the programming assignments, should be the name of the project and your name, e.g:

```
printf("Lab #5 - Joe Student\n");
```

Obviously, for the lab exercises, the output should include the names of both students in the lab group, e.g:

```
printf("Lab #5 - Joe Student and Jane Doe\n");
```

3 Sample Run

Your output should match the sample run below.

```
Lab #5 -- Mark Schwitzerlett and Jacobi
March 2, 2026

Enter number of judges (5 to 10): 6
Enter score for judge 1: 9.2
Enter score for judge 2: 8.7
Enter score for judge 3: 9.8
Enter score for judge 4: 9.1
Enter score for judge 5: 8.9
Enter score for judge 6: 9.5

Scores entered: 9.2 8.7 9.8 9.1 8.9 9.5
Highest score: 9.8
Lowest score: 8.7
Official score (drop high/low): 9.17


...Program finished with exit code 0
Press ENTER to exit console.
```

4 Deliverables

You should turn in a stand-alone, complete application program (your C source code, a single ".c" file) containing a `main()` function. All labs and projects this semester will be submitted in Gradescope (via a Canvas assignment).

Your file should be named Lab#5_FirstName_LastName.c

Due date: The lab is due at the end of your lab period, either Monday, March 2nd or Tuesday, March 3rd.

Be sure to save a copy of your .c file.